PAVITHRA RAJINDRA

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Education

Shiv Nadar University Chennai **2023 – 2027**
B.Tech in Computer Science and Engineering (Artificial Intelligence and Data Science) *CGPA: 8.77*
ML Vertical Head, Coding Club | Student Editor, SiNEWS (College Newsletter)

Indian Institute of Technology Madras (IIT-M - CODE) **2024 – 2028**
BS in Data Science and Applications (Online) *CGPA: 8.19*

Skills

Languages: Python, SQL, C/C++, Java, JavaScript
Coursework: Deep Learning (+Lab), Image & Video Processing (+Lab), Machine Learning Techniques (+Lab), NLP (+Lab), LLMs & Generative AI, Reinforcement Learning, Speech Technology; DL & GenAI, ML Foundations, ML Techniques, ML Practice
Deep Learning & Computer Vision: PyTorch, TensorFlow, OpenCV, CNNs, Diffusion Models (Stable Diffusion), Object Detection, Video Classification, Image Denoising, Transfer Learning
ML & Data: Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Model Evaluation & Benchmarking, Hyperparameter Tuning
Deployment & Tools: FastAPI, Flask, Git/GitHub, Weights & Biases, Hugging Face, Kaggle, Google Colab, Vercel
Web & Databases: React.js, Node.js, HTML/CSS, MySQL, SQLite, Google Firebase (Firestore)

Experience

Hexaware Technologies Ltd. **May '25 – Oct '25**
AI Internship | Hi-Tech & Professional Services Departments
- Developed and fine-tuned LLM-based solutions and internal AI tools for enterprise use cases, focusing on applied model customization and exploration of LLM capabilities.
- Conducted architectural analysis and fine-tuning experiments on large language models, documenting findings in a formal report for cross-team knowledge sharing.

Projects

Sketch-to-Product Image Generation with Structural Preservation **Jan '26 – Present**
Tech: Python, PyTorch, Stable Diffusion v1.5, T5, CLIP, Hugging Face
- Designing a modular four-stage **latent-diffusion pipeline** that **separates sketch structure from text-driven appearance** on a frozen Stable Diffusion v1.5: a CNN structure encoder feeds a lightweight ∼4M-parameter adapter (zero-initialized residuals into the UNet), while frozen T5 embeddings are projected into cross-attention tokens.

Image Denoising with Convolutional Autoencoders | Deep Learning Lab **Oct '25– Nov '25**
Tech: Python, TensorFlow/Keras, PSNR/SSIM
- Built and **compared four convolutional autoencoder variants** (shallow/deep × small/large filters) to denoise Gaussian-corrupted (σ=0.08) Tiny ImageNet images, using early stopping and learning-rate scheduling.
- **Benchmarked 11 ensemble combinations**: the **best** (shallow-large + deep-large) reached **27.90 dB PSNR / 0.907 SSIM** vs. 25.88 dB for the smallest model, with only a marginal gain over the best single model (27.76 dB).

Violence Identification in Video | Transfer Learning & Ensembles **Oct '25 – Oct '25**
Tech: Python, TensorFlow/Keras, CNN, MobileNetV2, ResNet50, Transfer Learning, Kaggle
- **Benchmarked a custom CNN** against frozen-backbone **MobileNetV2 and ResNet50** transfer-learning models for violent vs. non-violent video classification (frame-level, Real Life Violence and Non-Violence dataset), evaluated on accuracy, precision, recall, and F1.
- **Ensembling the CNN with MobileNetV2** lifted accuracy to **96.5%** (F1 0.966) from 94.1% and 94.8% for the individual models; the ResNet50 model lagged at 67.8%.

XenLearn – AI Learning Platform with Environmental Interaction (SNUC Hacks, 2nd Place) **Feb '25 – Feb '25**
Tech: Python, PyTorch, Torchvision, Faster R-CNN, OpenCV, Gemini API, Firebase, Streamlit
- Won 2nd place among 130+ competing teams by developing XenLearn, integrating **real-time object detection** and computer-vision-driven interaction into a **personalized, gamified learning** product.
- Collaborated with a 4-person cross-functional team to integrate the CV model, **AI-generated lesson content**, and a doubt-clearing chatbot into a single deployed Streamlit application, covering the pipeline from data handling to end-user product.

Video Dialogue Finder | Speech-to-Frame Search Pipeline **Aug '26 – Aug '26**
Tech: Python, PyTorch, Qwen3-ASR, Sentence-Transformers, RapidFuzz, OpenCV, FFmpeg, Pytest
- Built an end-to-end pipeline that finds the exact frame where a dialogue line is spoken: **Qwen3-ASR-1.7B** transcribes overlapping audio chunks with word-level millisecond timestamps, and hybrid **semantic embeddings** (MiniLM) + **fuzzy matching** (RapidFuzz) handles typos and paraphrased queries.
- Packaged it as a modular CLI with OpenCV frame extraction, JSON output (timestamp, frame number, confidence scores), and a six-module pytest suite that validated core functionality and edge cases.

Achievements & Leadership

- Competed in 6 IIT Madras-organized Kaggle/ML competitions (2025 – 2026) spanning data exploration, feature engineering, model experimentation, and performance analysis.
- ML Vertical Head, Coding Club, Shiv Nadar University – organized GitHub × Machine Learning workshops and mentored junior students on version control and ML workflows.
- Event Head, ModelWars, Invente '26 (Sep '26) – led a 12-member organising committee in end-to-end event planning and execution, driving competition design, technical coordination, and collaborative problem-solving for a gamified ML challenge.